Leakage-Aware Multi-Horizon Benchmarking of Compact Neural PV Forecasts Across Co-Located Technologies

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Photovoltaic (PV) forecasting benchmarks can favor a model through future-aware preprocessing, unequal eligible samples, or an undemanding reference rather than through forecasting skill. We evaluate four compact neural implementations on three co-located PV technologies at the DKA Solar Centre in Alice Springs. Each forecaster receives the same 17-channel, six-hour history and directly predicts 144 five-minute power values. The 1, 4, 8, and 12 h prefixes are evaluated on full-timeline and daylight scopes. Preprocessing is fitted on Train only, windows remain inside chronological splits, checkpoint selection uses Validation only, and every comparison shares forecast origins, labels, and point masks. For every primary combination, the lowest-RMSE compact neural implementation outperforms Last-value Persistence; the inverted-variate and depthwise implementations rank first in 12 and nine combinations, respectively. This Test-derived, post hoc best-of-four neural envelope is descriptive and is not a prespecified deployable model. Against a separately matched 24 h seasonal reference, Daily Persistence remains better than even this favorable envelope in 22 of 24 comparisons; the two envelope wins are Hanwha at the 1 h horizon under full-timeline and daylight evaluation. Rankings also vary by array, horizon, and scope. The findings show that causal preprocessing, elementwise sample matching, and a strong persistence challenge materially determine benchmark conclusions. The study is limited to co-located, history-only data and compact project implementations rather than full official architecture reproductions.

Keywords: photovoltaic power forecasting, multi-horizon forecasting, data leakage, persistence benchmark, reproducibility, co-located PV arrays

I. INTRODUCTION

Photovoltaic generation is variable at time scales relevant to distribution operation, storage scheduling, and reserve allocation. Forecast quality depends on the forecast horizon, temporal resolution, available observations, weather regime, and the definition of the target itself. Reviews therefore caution that numerical errors from different PV forecasting studies are rarely interchangeable without their complete data and evaluation context.[1–3] At the same time, rapid progress in recurrent, convolutional, Transformer-based, probabilistic, and weather-assisted forecasting has shifted attention toward increasingly elaborate architectures.[4–7]

Architecture is only one source of variation in a benchmark. Three evaluation choices can change the apparent conclusion before a model is fitted. First, imputation, anomaly detection, or scaling fitted on the full record transfers information from later periods into the representation of earlier periods. Second, methods may be compared on different eligible samples when missing targets, daylight filters, or lag requirements are handled separately. Third, a weak reference can make a learned model appear useful even when a simple operational alternative is better. These risks are especially consequential for PV data, where nighttime values are easy, missingness is structured, and daily periodicity is strong. General forecasting guidance and leakage audits emphasize temporal separation and explicit baselines, but published PV studies still differ in split definitions, persistence lags, missing-value rules, and eligible targets.[8–11]

The persistence question is not binary. Last-value Persistence extrapolates the power observed at the forecast origin and is a natural short-horizon continuity reference. Daily Persistence uses the power observed at the same clock time 24 h earlier and is a stronger seasonal reference when a valid lagged observation exists. The two baselines answer different questions and can use different raw supports. A fair comparison must therefore apply the chosen reference's actual valid point mask to every competing method rather than deleting unavailable points from the baseline alone.

Cross-system comparison introduces a second challenge. Different sites combine weather, climate, orientation, technology, capacity, sensor, and data-quality effects. The DKA Solar Centre in Alice Springs provides a useful controlled setting: several PV systems are co-located and share meteorological observations, while their power targets and module technologies differ.[12–14] Co-location does not establish cross-climate generality, but it reduces location-level confounding and permits a focused test of whether model rankings remain stable

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across targets exposed to broadly common weather.

This study addresses four research questions:

1. Under causal preprocessing and elementwise-matched evaluation, do compact neural implementations outperform Last-value Persistence?

2. Does that conclusion persist when the reference is changed to matched Daily Persistence?

3. How do rankings vary across PV arrays, 1–12 h horizons, and full-timeline versus daylight scopes?

4. Are accuracy differences accompanied by material differences in parameter count and inference cost?

The contributions are methodological and empirical rather than architectural. We (i) specify a leakage-aware multi-horizon protocol with Train-only preprocessing, split-local windows, Validation-only checkpoint selection, and identical evaluation support; (ii) compare three co-located PV technologies under shared meteorological context; (iii) demonstrate how Last-value and Daily Persistence lead to different but compatible scientific interpretations; and (iv) report seed variability, physical and normalized errors, sample counts, and hardware-specific efficiency for four compact neural implementations. The study does not propose a new neural network, reproduce every component of the architectures that inspired the implementations, or claim state-of-the-art or cross-site performance.

II. RELATED WORK

A. PV forecasting models and information availability

PV forecasting spans physical, statistical, machine-learning, and hybrid methods. Historical power and ground meteorology are commonly used for intra-hour and short intraday prediction; future numerical weather prediction (NWP), satellite products, and sky imagery become increasingly important as the horizon grows.[1,2] This distinction matters because a history-only model and an NWP-assisted model estimate different conditional quantities. Cross-Unet, for example, explicitly combines historical observations with forward weather fields, demonstrating the value of a richer forecast-time information set rather than providing a directly comparable error for a history-only task.[7]

Recent work has also broadened the scale and objectives of benchmarking. SolNet supplies open-source models across geographically distributed systems and emphasizes transfer across regions.[6] Markovics *et al.* compare classical and deep models using weather forecasts, extensive tuning, and multiple sites.[15] Dhingra *et al.* examine neural forecasts under data constraints and include uncertainty quantification, while related work by the same group studies quantile forecasts under data loss.[16,17] A 2025 review identifies standardization, generalization, and uncertainty as recurring open problems.[3] These studies are broader than the present work in geography, weather information, tuning, or uncertainty treatment. Our narrower estimand is the effect of evaluation discipline when causal history-only implementations are compared on exactly the same co-located samples.

Central Australian PV archives have already supported several forecasting studies. Mahmud *et al.* used machine learning for power forecasting at Alice Springs, Huang *et al.* evaluated a neural-ODE-based architecture on three Alice Springs arrays, and Mansour *et al.* benchmarked six machine-learning models against Persistence on high-resolution Yulara data under seasonal and stress conditions.[14,18,19] These studies establish the relevance of public desert-PV data but do not make the current evaluation contribution redundant. Data availability, split definitions, persistence lags, model inputs, and eligible samples differ across studies; consequently, reported numerical errors are not directly comparable.

B. Compact time-series mechanisms

Long short-term memory networks and temporal convolutional networks are established sequence models.[20,21] Transformer variants introduced efficient attention, decomposition, frequency-domain structure, patch tokenization, and inverted variate tokens.[22–27] ModernTCN revisited convolutional time-series modeling with large kernels and modern block design.[28] These papers define architecture families, not the exact compact implementations evaluated here.

Our four forecasters retain one recognizable mechanism from those families under a common input-output budget: recurrent trajectory decoding, variate tokenization, joint temporal patching, or depthwise temporal convolution. They intentionally omit or modify substantial components. The comparison therefore concerns implemented mechanisms under one controlled protocol. It must not be interpreted as a leaderboard for official iTransformer, PatchTST, or ModernTCN releases.

C. Persistence, leakage, and reproducible evaluation

Forecasting benchmarks require naive references whose information sets are explicit.[29,30] Persistence is particularly competitive for solar generation because power is smooth over minutes and approximately periodic over days. Skill relative to persistence communicates whether a forecast adds value beyond that structure, but only if the reference and model share targets and masks. Daily lag availability can otherwise create a hidden sample-selection effect.

Temporal leakage can enter through preprocessing, windows, or model selection. Random cross-validation is inappropriate for many dependent series, and globally fitted transformations can make a nominally chronological evaluation transductive.[9,10] Reproducibility further requires definitions of split boundaries, target timestamps, inverse scaling, checkpoint selection, and metric aggregation. Three random seeds cannot establish universal statistical superiority, but reporting them prevents a favorable initialization from being silently selected.

Table I positions the present study against recent direct competitors. The distinctive combination is not any single architecture; it is the simultaneous control of causal preprocessing, point-level eligible samples, weak and strong persistence references, multiple horizons, two evaluation scopes, and unified inference timing on co-located technologies.

III. DATA AND EXPERIMENTAL PROTOCOL

A. Co-located PV systems and temporal splits

The DKA Solar Centre in Alice Springs, Australia, provides five-minute PV and meteorological measurements through its official data service and system metadata pages (accessed August 28, 2026).[12] We use Site 17 Sanyo HIT-210NKHE5 hybrid-silicon modules (6.30 kW rated DC), Site 25 Hanwha HSL 60S polycrystalline-silicon modules (5.83 kW DC), and Site 38 Q CELLS Q.PEAK-G4.1 monocrystalline-silicon modules (5.90 kW DC). The target is Active Power in kW AC. Official AC capacities were unavailable, so DC ratings are reported descriptively and never used to normalize AC prediction error.

The aligned systems share six meteorological fields: ambient temperature, relative humidity, global and diffuse horizontal radiation, and global and diffuse tilted radiation. Each array retains its own Active Power, Performance Ratio, and missingness. Co-location controls much of the weather and site context but does not isolate module technology as a randomized causal treatment. Orientation, balance-of-system behavior, and sensing can also contribute to differences.

Train covers April 1–July 15, 2018; Validation covers July 16–August 7; and Test covers August 8–31, all on Australian Central Standard Time. Table II reports horizon-specific Test origins. Counts decline with horizon because a longer valid prefix is required. Adjacent origins overlap and are therefore repeated operational forecast instances, not independent statistical experiments.

B. Causal preprocessing and window construction

Raw records are reindexed to a regular five-minute coordinate without deleting nighttime rows. A missing timestamp remains explicitly missing rather than becoming adjacent to a distant observation. The eight causal numerical inputs are Performance Ratio, six weather fields, and historical Active Power. Each numerical channel has a missing indicator, and one Isolation Forest indicator is added, giving 17 channels. There are no cyclical time channels. The Isolation Forest, five-neighbor imputer, feature scaler, and target scaler are fitted on Train only and then applied unchanged to Validation and Test.[31,32]

Each input contains 72 consecutive positions on the regular five-minute coordinate (6 h). The forecast origin is the final input timestamp, and the first target is exactly five minutes later. Future Active Power never enters the input. Windows are constructed independently inside each chronological split and never cross split boundaries. Missing input observations remain on the regular coordinate and are represented by missingness masks plus Train-only imputation; they are not treated as breaks in time. Target eligibility is evaluated independently for each requested horizon. Figure 1 summarizes the design.

All models are trained to predict the complete H144 trajectory, and checkpoint selection minimizes one global masked Validation MSE computed as total squared error divided by total valid target count. Batch means are not weighted equally. Test is excluded from preprocessing fitting, optimization, early stopping, and checkpoint selection. Primary Test evaluation uses horizon-specific eligibility: H12 requires a valid L72 history and H12 target prefix, whereas H144 requires the complete target. A complete-H144 common-origin analysis is retained as sensitivity analysis in the Supplementary Material.

C. Architecture identity and training configuration

Table III states what each implementation does and, equally important, what it does not reproduce. All map an input of shape 72×17 to a 144-value trajectory. The Discrete recurrent decoder is project-defined: a multi-dilation convolutional branch and a one-layer Transformer branch are gated, summarized by a bidirectional GRU, and decoded through a GRUCell. The Inverted-variate Transformer treats each of the 17 variables as a token by projecting its 72-point history, but uses one compact encoder and a joint flattened output rather than the full official iTransformer design. The Joint-patch Transformer forms overlapping multivariate patches; unlike PatchTST, it is not channel-independent and does not implement the full official normalization and head design. The Depthwise convolutional TCN stacks kernel-5 depthwise and pointwise convolutions; it omits ModernTCN's full patching, large-kernel, block, and reparameterization design.

Each implementation uses seeds 42, 43, and 44. The shared optimization budget is AdamW with learning rate 0.001, weight decay 10^{-5}, batch size 256, maximum 25 epochs, gradient clipping at 1.0, and early-stopping patience 5.[33] No Test metric selects a seed, architecture, epoch, horizon, or threshold. The comparison holds optimization budget and information set constant while al-

TABLE I. Positioning against directly related PV forecasting studies. "Future information" refers to exogenous variables available after the forecast origin but issued beforehand.

Study	Year	Systems	Forecast-time information	Horizons	Scope / reference	Main distinction
SolNet[6]	2025	Global sites	History; transfer	Multiple	Open deep models	Geographic breadth; no co-located mask audit
Di Leo *et al.*[3]	2025	Review	Multiple regimes	Multiple	Deterministic / probabilistic	Synthesis of standardization and uncertainty
Markovics *et al.*[15]	2026	Multiple plants	NWP weather forecasts	Multiple	Classical/deep; skill	Broader sites, models, tuning, and information
Dhingra *et al.*[16]	2026	Two PV sites	Historical irradiance	Short-term	Neural; uncertainty	Missing-data stress and probabilistic quality
Cross-Unet[7]	2026	Multiple PV tasks	History + future weather	Multiple	Dual-stream neural	Richer future information and dedicated fusion
Mahmud *et al.*[14]	2021	Alice Springs	Historical PV/weather	Short/long	ML; conventional	Same facility; different task and sample rules
Mansour *et al.*[19]	2026	Yulara system	Historical PV, weather, engineered lags	5–15 min resolution	Six ML models; Persistence	Seasonal/stress benchmark; different site and estimand
This study	2026	Three co-located arrays	6 h causal history only	1, 4, 8, 12 h	Compact neural; two Persistence lags	Elementwise matching across technologies and scopes

TABLE II. Dataset and primary Test support. Counts are identical for each neural implementation and Last-value Persistence at a given array and horizon.

Array (site)	Technology	Rated DC	Target	H12 origins	H48 origins	H96 origins	H144 origins
Sanyo (17)	HIT hybrid silicon	6.30 kW	Active Power (kW AC)	6,589	5,905	5,024	4,160
Hanwha (25)	Polycrystalline silicon	5.83 kW	Active Power (kW AC)	6,625	6,049	5,281	4,521
Qcells (38)	Monocrystalline silicon	5.90 kW	Active Power (kW AC)	6,463	5,491	4,227	2,996

lowing parameter count to vary naturally with the mechanism.

D. Persistence references and evaluation scopes

Last-value Persistence repeats the most recent observed power at the forecast origin across H144. It shares every primary origin, label, and point mask with each neural forecast. Daily Persistence retrieves the real power at each target timestamp minus 24 h (288 five-minute steps), without interpolation. Because some lagged points are unavailable, the Daily-valid point mask is intersected with the target mask and then applied identically to Daily Persistence, Last-value Persistence, and every neural prediction. This supplementary comparison has its own matched support and is not mixed into the primary rank table.

The full-timeline scope includes all valid targets. The daylight scope is defined after prediction as true target power above 1% of the corresponding array's Train maximum. It is an evaluation mask, not an input or operational daylight detector. At Qcells H12, for example, 6,463 origins yield 77,556 full target points and 36,504 daylight points (47.1%).

E. Metrics and computational measurement

For valid points $i = 1, \ldots, n$, we compute

$$\mathrm{RMSE} = \sqrt{n^{-1} \sum_i (\hat{y}_i - y_i)^2}, \tag{1}$$

$$\mathrm{MAE} = n^{-1} \sum_i |\hat{y}_i - y_i|, \tag{2}$$

$$\mathrm{Bias} = n^{-1} \sum_i (\hat{y}_i - y_i), \tag{3}$$

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{4}$$

Negative R^2 is retained. Because AC nameplate capacity is unknown, normalized RMSE uses the Train target range,

$$\text{range-nRMSE} = \frac{\mathrm{RMSE}}{y_{\max,\mathrm{Train}} - y_{\min,\mathrm{Train}}}. \tag{5}$$

The denominator is never fitted on Test. RMSE skill relative to reference b is $1 - \mathrm{RMSE}_{model}/\mathrm{RMSE}_b$; positive values favor the model. Neural values are means with sample standard deviations over three seeds. The 24 array–horizon–scope combinations are dependent summaries, not 24 independent trials, so we do not attach naive p-values. We also compute a post hoc best-of-four neural envelope by taking, within each Test combination, the smallest three-seed mean RMSE among the four implementations. This descriptive upper bound favors the

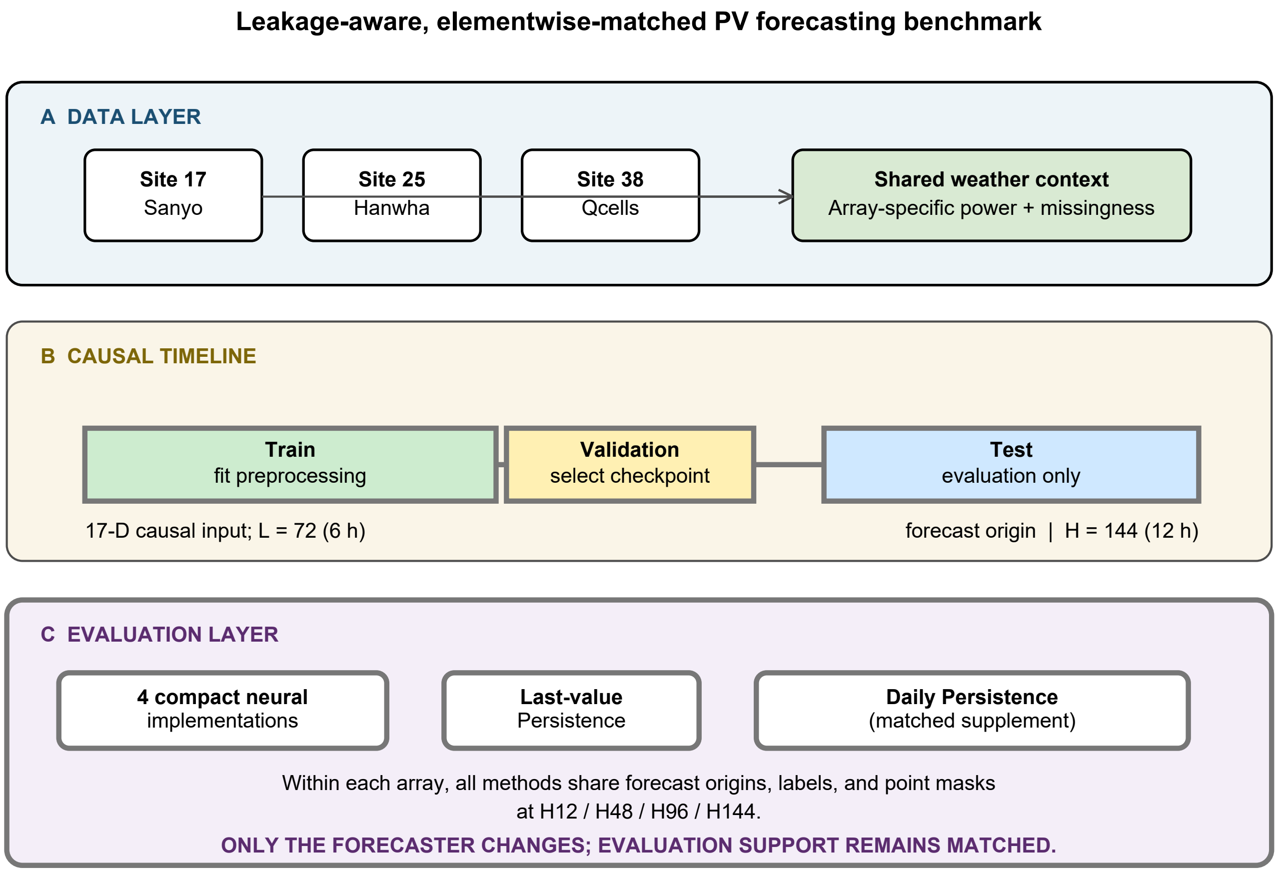

FIG. 1. Three-layer benchmark design. Co-located arrays share meteorological context but retain array-specific power and missingness. Train-only transformations and split-local windows preserve causality. Within each array, neural and persistence forecasts use the same origins, labels, and point masks.

TABLE III. Architecture identity and permitted interpretation. Parameter counts and synchronized batch-one latency are for the 17-channel H144 task.

Manuscript name	Retained mechanism / omitted components	Parameters / latency	Permitted interpretation
Discrete recurrent decoder	Multi-dilation convolution, Transformer, gated fusion, GRU; project-defined recurrent decoder	99,362 32.204 ms	Implemented recurrent decoding mechanism only
Inverted-variate Transformer	Variate tokens inspired by iTransformer[27]; one encoder and flattened head	194,960 0.558 ms	Compact variate-token; not official iTransformer
Joint-patch Transformer	Overlapping joint patches inspired by PatchTST[26]; no channel independence or full official head	148,112 0.535 ms	Compact joint-patch; not official PatchTST
Depthwise convolutional TCN	Kernel-5 depthwise/pointwise convolutions; no full ModernTCN patching or large-kernel blocks[28]	683,024 0.706 ms	Compact depthwise TCN; not official ModernTCN

neural group and is neither a prespecified model choice nor a deployable forecaster.

Inference is measured in float32 on one NVIDIA GeForce RTX 3060 Laptop GPU under Windows and PyTorch. Batch-one latency uses evaluation mode, warm-up, synchronized repetitions, and input shape $[1, 72, 17]$. Throughput uses batch 256. Timing excludes model loading, disk I/O, data loading, and host-to-device transfer. The recorded field is Peak GPU memory (MiB). These values describe the stated hardware rather than general deployment speed.

F. AI-assisted workflow disclosure draft

Author confirmation required. OpenAI Codex (provider: OpenAI; exact application version and model designation to be confirmed by the authors) assisted with code review, organization of evidence-recalculation checks, extraction of bibliographic metadata, and language editing. It was used to improve consistency and traceability, not to create or alter source observations, trained weights, or frozen numerical results. The authors executed the analysis and verification scripts, checked quantitative outputs against saved artifacts, checked bibliographic claims against source publications, and retain responsibility for the study. AI tools are not authors. This draft must be confirmed or revised by all authors before submission.

IV. RESULTS

A. Sample support and evaluation integrity

The evidence contains 36 completed neural runs (four implementations, three arrays, and three seeds). Within an array, labels, target-valid masks, forecast origins, and target starts are elementwise identical across all runs. Every saved prediction has shape $N \times 144$ and finite values. An independent NumPy/Pandas implementation re-derived the principal masks, metrics, ranks, persistence comparisons, and Qcells counts from saved arrays without importing the production metric functions. All 4,414 numerical comparisons agreed within floating-point tolerance.

Horizon-specific eligibility materially preserves short-horizon evidence. For Qcells, H12 uses 6,463 origins whereas H144 uses 2,996. This is not an advantage granted to one method: every H12 forecaster and Last-value Persistence uses the same 6,463 origins and 77,556 targets. Full counts for all horizons and scopes appear in the Supplementary Material.

B. Neural implementations versus Last-value Persistence

Across the 24 primary combinations, the Inverted-variate Transformer records 12 lowest-RMSE outcomes, the Depthwise TCN nine, the Joint-patch Transformer two, and the Discrete recurrent decoder one; Last-value Persistence records none. The average ranks are 1.875, 2.167, 2.458, 3.667, and 4.833, respectively. The arithmetic mean RMSE skills relative to Last-value Persistence are 0.549, 0.515, 0.493, and 0.399 for the four neural implementations in the same order. These ratio means are not an arithmetic mean of absolute errors.

Table IV shows H12 and H144 endpoints; complete H12/H48/H96/H144 results are given in the supplement. At full H144, the post hoc neural-envelope RMSE is 0.776 kW for Sanyo, 0.743 kW for Hanwha, and 0.789 kW for Qcells, compared with Last-value values of 2.729, 2.561, and 2.418 kW. At Qcells H12, Last-value RMSE is 0.471 kW; the Inverted-variate implementation reaches 0.327 kW, whereas two compact implementations remain worse than Last-value. Thus, “all neural models beat Last-value” is not a valid per-combination statement even though the post hoc best-of-four neural envelope is lower in each primary comparison.

C. Strong-baseline reversal under Daily Persistence

Figure 2 contrasts the post hoc best-of-four neural envelope against two causal references. Its RMSE ratio to Last-value Persistence is below one in all 24 primary comparisons. On the elementwise Daily-matched mask, the ratio to Daily Persistence exceeds one in 22 comparisons. The two exceptions are Hanwha H12: full-timeline RMSE is 0.176 kW for the Depthwise TCN versus 0.185 kW for Daily Persistence, and daylight RMSE is 0.231 versus 0.273 kW. Because the envelope selects its member after Test results are available, it is a favorable descriptive bound rather than a deployable model.

The findings are compatible rather than contradictory. Last-value Persistence measures improvement beyond local continuity at the forecast origin. Daily Persistence measures improvement beyond a strong 24 h seasonal pattern. Neural training is effective relative to the first reference, while its additional value relative to the second is limited to short-horizon Hanwha forecasts in the present evaluation. The result restricts claims of practical advantage without implying that optimization failed.

D. Dependence on technology, horizon, and daylight scope

Figure 3 shows that errors grow with horizon and rise when near-zero targets are excluded. At H144, the lowest neural Train-range nRMSE changes from 0.128 to 0.175 for Sanyo, 0.128 to 0.178 for Hanwha, and 0.131 to 0.191 for Qcells when moving from full timeline to daylight. Scope also changes model identity: Sanyo H12 favors the Discrete recurrent decoder on the full timeline but the Depthwise TCN in daylight.

Array-specific ranking should not be reduced to module technology alone. The Inverted-variate implementation leads most Qcells prefixes and Sanyo H96/H144, whereas the Depthwise TCN leads most Hanwha settings. Joint-patch leads both Sanyo H48 scopes, and Discrete recurrent leads Sanyo H12 full timeline. Colocation makes these differences observable under common weather, but electrical design, orientation, sensors, missingness, and target distribution remain potential explanations alongside module technology.

E. Accuracy-efficiency trade-offs

Table III and Fig. 4 show that accuracy, size, and runtime do not have one ordering. The Inverted-variate implementation has the lowest macro Train-range nRMSE. Joint-patch has the lowest batch-one latency and highest

TABLE IV. Primary Test Train-range nRMSE on horizon-specific valid origins. The envelope is the post hoc minimum three-seed mean among the four compact implementations and is a descriptive upper bound, not a prespecified model; the complete model-by-seed table is in the Supplementary Material.

Array	Scope	Horizon	Envelope member	Envelope nRMSE	Last-value nRMSE	RMSE skill
Sanyo	Full	H12	Discrete recurrent	0.0342	0.0766	0.553
Sanyo	Full	H144	Inverted-variate	0.1280	0.4506	0.716
Sanyo	Daylight	H12	Depthwise TCN	0.0480	0.1101	0.564
Sanyo	Daylight	H144	Inverted-variate	0.1745	0.6056	0.712
Hanwha	Full	H12	Depthwise TCN	0.0301	0.0745	0.595
Hanwha	Full	H144	Depthwise TCN	0.1275	0.4394	0.710
Hanwha	Daylight	H12	Depthwise TCN	0.0396	0.1078	0.633
Hanwha	Daylight	H144	Depthwise TCN	0.1782	0.5614	0.683
Qcells	Full	H12	Inverted-variate	0.0542	0.0780	0.305
Qcells	Full	H144	Depthwise TCN	0.1307	0.4005	0.674
Qcells	Daylight	H12	Inverted-variate	0.0756	0.1129	0.330
Qcells	Daylight	H144	Depthwise TCN	0.1909	0.6068	0.685

throughput. Discrete recurrent has the fewest parameters but is much slower because 144 recurrent decoding steps serialize work. Depthwise TCN is largest and does not dominate the accuracy frontier.

The measured forward pass omits preprocessing, transfer, and data ingestion. An operational assessment would also consider batching policy, power consumption, reliability, and the target device. The present measurement isolates implementation overhead under one reproducible environment.

F. Sensitivity analyses

Restricting every prefix to complete-H144 origins lowers some errors because it selects longer uninterrupted target periods. That analysis is reported in the supplement but is not the primary result: a valid one-hour forecast should not be discarded because hours 2–12 contain missing targets. The complete-H144 sensitivity preserves a common origin set across horizons, while the horizon-specific primary analysis preserves the estimand at each requested horizon. Neither design is hidden.

The rank heat map in the supplement shows the full 24-combination pattern. It confirms that no compact neural implementation has a constant rank across arrays, horizons, and scopes. Per-seed tables similarly show optimization variability; three seeds are sufficient to prevent best-seed reporting but not to justify strong inferential claims.

V. DISCUSSION

A. Why the persistence references lead to different conclusions

Last-value Persistence exploits continuity. At one hour, a model can improve by learning recent gradients, irradiance response, and mean reversion; at 12 h, holding the origin value is physically implausible across day-night transitions. Daily Persistence instead places the forecast near the expected diurnal level for each future timestamp. During a short, seasonally narrow Test period, that pattern is difficult for a history-only neural model to beat even when the latter predicts a coherent H144 trajectory.

The strong Daily result is not caused by allowing it an easier sample. Its finite-lag mask is imposed on every method in the supplementary analysis. Conversely, Daily Persistence is not included in the primary ranks because its raw support differs before matching. Reporting both comparisons preserves two estimands: skill beyond immediate continuity and skill beyond 24 h recurrence.

The two Hanwha H12 exceptions are informative but narrow. They indicate that recent multivariate history can add short-horizon information beyond yesterday's value for this array. They do not establish a universal technology effect, and 24 combinations are not independent replications. A future study would need a predeclared external time block and more sites before using these exceptions to motivate deployment decisions.

B. Information limits of history-only deterministic forecasts

Historical power and irradiance identify the present state and recent variability, but they do not uniquely determine future cloud motion. With squared-error training, unresolved futures are averaged, favoring smoother trajectories. This mechanism is consistent with the increasing error at long horizons and with the advantage of a daily seasonal baseline. It does not prove that future weather would solve every error, but it defines the information boundary of the current task.

Recent literature illustrates complementary routes beyond that boundary. Cross-Unet uses forward weather in a dedicated historical/future fusion architecture, and Markovics *et al.* benchmark weather-based models at

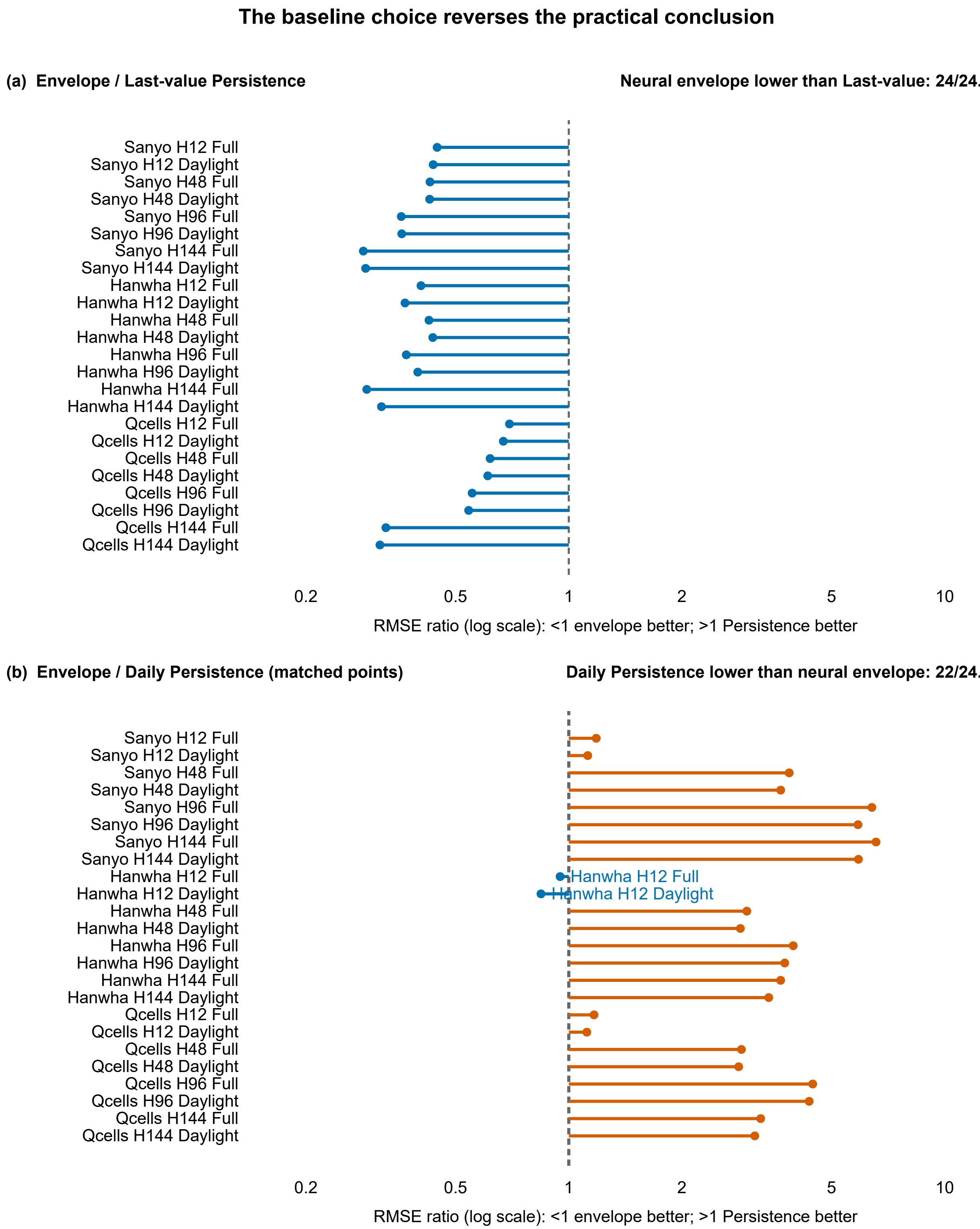

FIG. 2. Baseline reversal across the same 24 array–horizon–scope orderings. Each point is the RMSE of the post hoc best-of-four neural envelope divided by the reference RMSE. Panel A uses Last-value Persistence on primary support; panel B uses Daily Persistence on its elementwise-matched support. Ratios below one favor the envelope, one is a tie, and ratios above one favor Persistence.

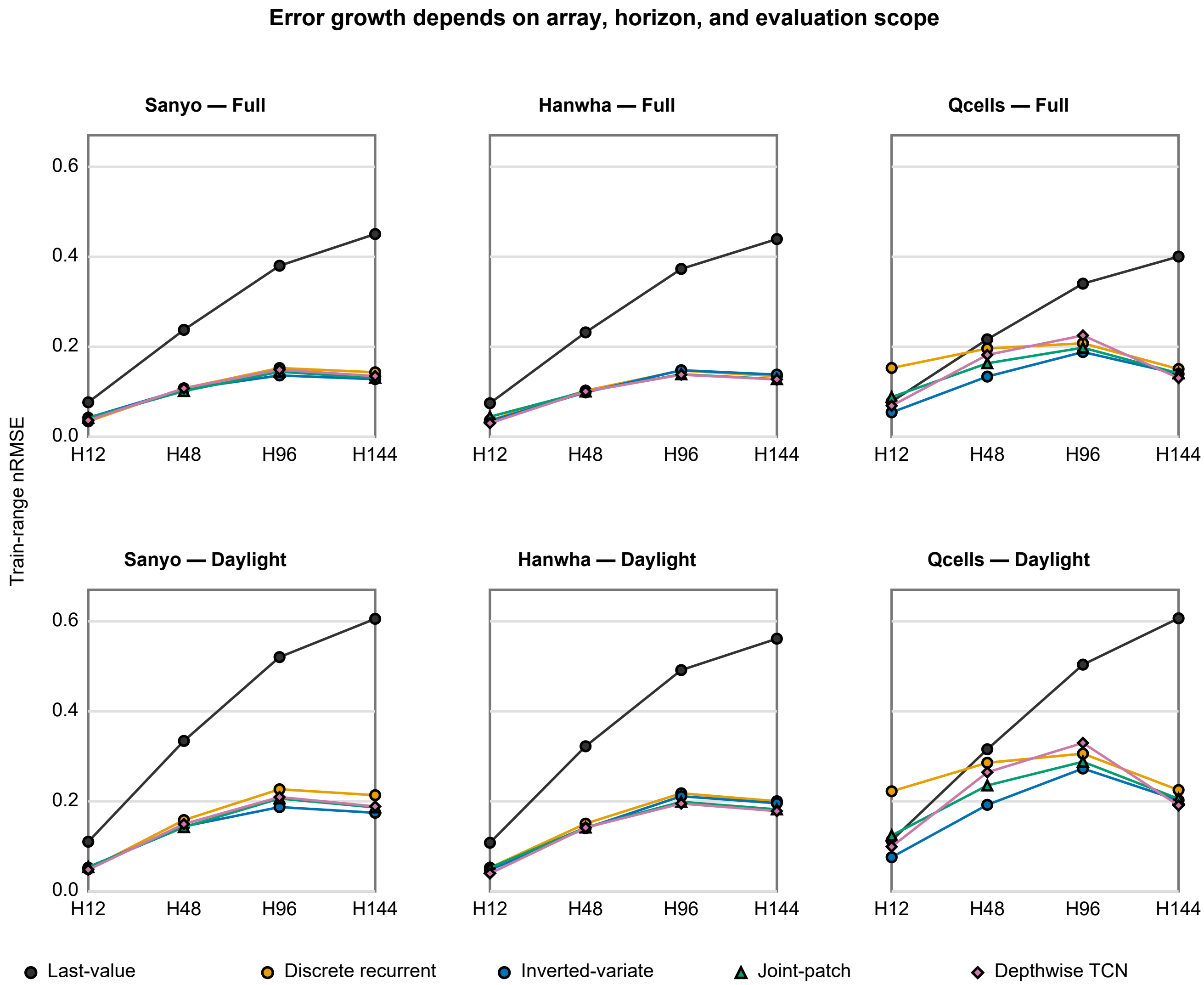

FIG. 3. Primary Train-range nRMSE across horizons, arrays, and evaluation scopes. Rows separate full-timeline and daylight results, and columns separate the three co-located arrays. Every point uses horizon-specific origins and masks shared by all methods in its panel.

broader scale.[7,15] Dhingra *et al.* address uncertainty and missing-data stress rather than only point accuracy.[16] SolNet emphasizes geographic transfer.[6] The present benchmark does not compete with those objectives. Its contribution is to show that, even before new architecture or information is considered, the evaluation support and reference forecast can reverse the practical conclusion.

C. Implications for benchmark design

Five practices follow from the evidence. Preprocessing fit boundaries should be stated alongside date splits. Windows should be audited at the timestamp level, not inferred from row order. A causal target history available to persistence should also be available to a learned history model. Each horizon should report forecast-origin and valid-target counts. Finally, baseline-specific missingness should be handled through an intersection mask shared by every method.

Full-timeline and daylight metrics answer different questions. Full-timeline evaluation reflects all scheduled predictions, including correctly near-zero production. Daylight evaluation isolates production-period difficulty but uses true future power and is therefore descriptive. Defining daylight from each prediction would make support model-dependent; our target-based mask is fixed after prediction and shared.

Normalization requires similar care. Dividing AC error by a DC module rating mixes quantities affected by inverter capacity and losses. We use a Train-only target range because official AC capacities are unknown. This

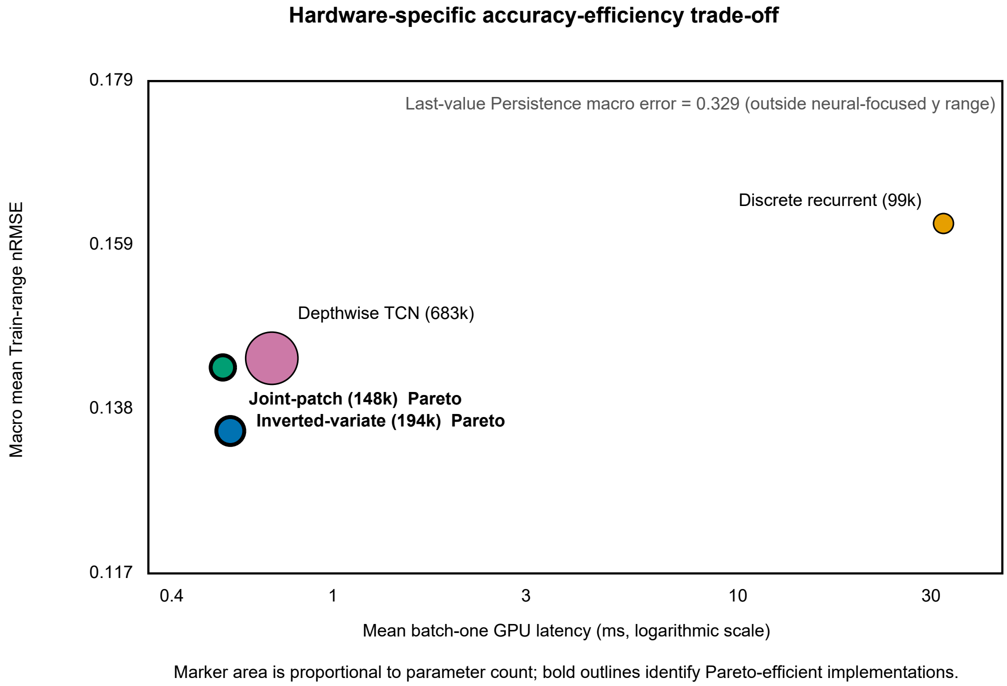

FIG. 4. Accuracy-efficiency trade-off on one RTX 3060 Laptop GPU. The vertical coordinate is the unweighted mean primary Train-range nRMSE over 24 combinations; the horizontal coordinate is synchronized batch-one latency. Marker area is proportional to parameter count, and bold outlines indicate Pareto-efficient implementations. Hardware-specific latency is not a deployment guarantee.

denominator is leakage-safe but dataset-dependent and must not be treated as a universal capacity-normalized metric.

D. Scope of the cross-technology evidence

The three arrays experience broadly common weather, which reduces one source of confounding compared with geographically separated systems. Their rankings nevertheless differ. This is evidence that conclusions from one target do not automatically transfer to another co-located target. It is not evidence that cell technology alone causes the difference. The study estimates forecasting behavior for three documented systems, not an average treatment effect for hybrid, polycrystalline, and monocrystalline technologies.

This scope distinguishes the work from global and multi-site benchmarks. SolNet and the Markovics benchmark are stronger for geographic breadth and transfer or NWP-based comparisons.[6,15] The current design is stronger only for its tightly matched within-site evaluation question. Numerical scores should not be compared across these papers because information availability, sampling, and splits differ.

VI. LIMITATIONS

The experiment covers one Alice Springs facility and April–August 2018. Three arrays do not represent all products within their module categories, and shared weather does not imply shared electrical behavior. No future NWP, satellite imagery, sky imagery, or probabilistic output is used. The estimand is therefore deterministic history-only forecasting, not the best achievable operational forecast.

The four networks are compact project implementations. They isolate broad mechanisms under a common budget but are not official full iTransformer, PatchTST, or ModernTCN reproductions. Results cannot be used to rank those published architectures. Only three random seeds are available, and overlapping forecast origins are dependent. Means and sample standard deviations describe initialization variability; no claim of statistical significance is made.

Official AC capacities are unavailable, so Train-range normalization is used. The daylight mask depends on true target power and is not deployable. Daily Persistence has a smaller raw lag-valid set than the primary comparison, although all methods are exactly matched within its supplementary analysis. Test was excluded from preprocessing fitting, training, and checkpoint selection, but it was inspected repeatedly during project development and is not an untouched external confirmation set. Replication should reserve a new time period before any additional model decisions.

Computational results come from one laptop GPU and omit the surrounding pipeline. They describe relative forward-pass behavior in this environment, not embedded-device suitability. Public reproduction also depends on users obtaining the source DKASC data under provider terms; large checkpoints and prediction arrays are not part of the planned public release.

VII. CONCLUSION

Under Train-only preprocessing, causal split-local windows, and elementwise-matched evaluation, every primary comparison contains at least one compact neural implementation with lower RMSE than Last-value Persistence. The Inverted-variate implementation records 12 of 24 primary RMSE wins, the Depthwise TCN nine, and rankings change with array, horizon, and daylight scope. The stronger matched Daily Persistence reference is lower than the favorable post hoc neural envelope in 22 of 24 comparisons; envelope wins are limited to Hanwha H12 full-timeline and daylight evaluation.

Accordingly, the main result is not a universal architecture advantage. It is that credible PV benchmarks require causal preprocessing, common evaluation support, explicit horizon and scope definitions, and persistence references that reflect both local continuity and daily seasonality. These controls reveal where compact neural forecasts add value and where a simple reference remains stronger. The evidence supports more disciplined evaluation practice for co-located history-only PV forecasting, while broader operational claims require future exogenous information, additional sites, and an independent temporal confirmation set.

SUPPLEMENTARY MATERIAL

See the Supplementary Material for full model configurations, architecture-identity details, complete per-seed metrics, all evaluation counts, Daily-matched results, complete-H144 sensitivity, the rank heat map, independent verification summary, efficiency protocol, and metric definitions.

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CONFLICT OF INTEREST

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ETHICS APPROVAL

Author confirmation required. Draft: This study analyzes publicly accessible photovoltaic and meteorological records and involves no human participants or animals; ethics approval was not applicable.

AUTHOR CONTRIBUTIONS

Author confirmation required. Draft: Jiangkun Zhu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – Original Draft, Visualization. Mengling Yang: Validation, Writing – Review & Editing. Zhicong Chen: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – Review & Editing. Lijun Wu: Supervision, Resources, Project administration, Funding acquisition, Writing – Review & Editing.

DECLARATION OF AI-ASSISTED TOOLS

Author confirmation required. The draft disclosure and verification responsibilities are stated in Methods. All authors must confirm the exact tool/version record and final wording before submission.

DATA AVAILABILITY

Author confirmation required. Draft: The source PV and meteorological observations are available from the DKA Solar Centre data service and official metadata pages.[12] The original data remain subject to provider terms. Aggregate evidence, split definitions, and reproduction instructions are planned for a dedicated code release.

CODE AVAILABILITY

Author confirmation required. Draft: A dedicated public release of evaluation code, aggregate metrics, figure generation, and ordinary protocol tests is planned after manuscript approval and license selection. Raw data, checkpoints, and prediction arrays will not be redistributed. The final repository URL must be confirmed before submission.

[1] J. Antoñanzas, N. Osorio, R. Escobar, R. Urraca, F. J. Martinez-de Pison, and F. Antonanzas-Torres, "Review of photovoltaic power forecasting," Solar Energy **136**, 78–111 (2016).

[2] A. Mellit, A. M. Pavan, and V. Lughi, "Deep learning neural networks for short-term photovoltaic power forecasting," Renewable Energy **172**, 276–288 (2021).

[3] P. Di Leo, A. Ciocia, G. Malgaroli, and F. Spertino, "Advancements and challenges in photovoltaic power forecasting: A comprehensive review," Energies **18**, 2108 (2025).

[4] H. Hewamalage, C. Bergmeir, and K. Bandara, "Recurrent neural networks for time series forecasting: Current status and future directions," International Journal of Forecasting **37**, 388–427 (2021).

[5] K. Benidis, S. S. Rangapuram, V. Flunkert, Y. Wang, D. Maddix, C. Turkmen, J. Gasthaus, and T. Januschowski, "Deep learning for time series forecasting: Tutorial and literature survey," ACM Computing Surveys **55**, 1–36 (2022).

[6] J. Depoortere, J. Driesen, J. A. K. Suykens, and H. S. Kazmi, "SolNet: Open-source deep learning models for photovoltaic power forecasting across the globe," International Journal of Forecasting **41**, 1223–1236 (2025).

[7] Y. Zhang, Y. Zhang, Z. Dai, and R. Zhang, "Rethinking the use of deep learning methods for photovoltaic power forecasting," Nature Communications (2026), 10.1038/s41467-026-73817-3.

[8] L. J. Tashman, "Out-of-sample tests of forecasting accuracy: An analysis and review," International Journal of Forecasting **16**, 437–450 (2000).

[9] C. Bergmeir and J. M. Benítez, "On the use of cross-validation for time series predictor evaluation," Information Sciences **191**, 192–213 (2012).

[10] H. Hewamalage, K. Ackermann, and C. Bergmeir, "Forecast evaluation for data scientists: Common pitfalls and best practices," Data Mining and Knowledge Discovery **37**, 788–832 (2023).

[11] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns **4**, 100804 (2023).

[12] Desert Knowledge Australia Solar Centre, "Data download: Dkasc, alice springs," (2026), official data service and system metadata; accessed 28 August 2026.

[13] K. Wang, X. Qi, and H. Liu, "A comparison of day-ahead photovoltaic power forecasting models based on deep learning neural network," Applied Energy **251**, 113315 (2019).

[14] K. Mahmud, S. Azam, A. Karim, S. M. Zobaed, B. Shanmugam, and D. Mathur, "Machine learning based PV power generation forecasting in alice springs," IEEE Access **9**, 46117–46128 (2021).

[15] D. Markovics, G. B. Békési, and M. J. Mayer, "A unified benchmark of deep learning and classical machine learning models for weather-based solar power forecasting: Balancing complexity and skill," Energy and AI **25**, 100833 (2026).

[16] S. Dhingra, G. Gruosso, and G. Storti Gajani, "Short-term PV power forecasting under real-world data constraints: A benchmark study of neural networks with uncertainty quantification," Neural Computing and Applications **38**, 626 (2026).

[17] S. Dhingra, G. Gruosso, and G. Storti Gajani, "Quantile-based short-term probabilistic forecasting of solar PV power under data loss," in *51st Annual Conference of the IEEE Industrial Electronics Society (IECON)* (2025).

[18] S. Huang, Q. Zhou, J. Shen, H. Zhou, and B. Yong, "Multistage spatio-temporal attention network based on NODE for short-term PV power forecasting," Energy **290**, 130308 (2024).

[19] H. S. E. Mansour, A. S. Mohamed, H. M. H. Farh, A.-W. Ibrahim, A. M. Al-Shaalan, A. B. Kunya, and H. S. Elnashar, "High-resolution photovoltaic power forecasting using machine learning models under seasonal and stress conditions," Scientific Reports **16**, 19029 (2026).

[20] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation **9**, 1735–1780 (1997).

[21] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," arXiv preprint arXiv:1803.01271 (2018).

[22] A. Vaswani, N. Shazeer, N. Parmar, *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).

[23] H. Zhou, S. Zhang, J. Peng, *et al.*, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35 (2021) pp. 11106–11115.

[24] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Advances in Neural Information Processing Systems*, Vol. 34 (2021) pp. 22419–22430.

[25] T. Zhou, Z. Ma, Q. Wen, *et al.*, "FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting," in *Proceedings of the 39th International Conference on Machine Learning*, PMLR, Vol. 162 (2022) pp. 27268–27286.

[26] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *International Conference on Learning Representations* (2023).

[27] Y. Liu, T. Hu, H. Zhang, *et al.*, "iTransformer: Inverted transformers are effective for time series forecasting," in *International Conference on Learning Representations* (2024).

[28] D. Luo and X. Wang, "ModernTCN: A modern pure convolution structure for general time series analysis," in *International Conference on Learning Representations* (2024).

[29] R. J. Hyndman and A. B. Koehler, "Another look at measures of forecast accuracy," International Journal of Forecasting **22**, 679–688 (2006).

[30] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The m4 competition: 100,000 time series and 61 forecasting methods," International Journal of Forecasting **36**, 54–74 (2020).

[31] O. Troyanskaya, M. Cantor, G. Sherlock, *et al.*, "Missing value estimation methods for dna microarrays," Bioinformatics **17**, 520–525 (2001).

[32] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining* (2008) pp. 413–422.

[33] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations* (2019).